

Algebra_Lineal06_06_26

June 6, 2026

```
[1]: import numpy as np
```

```
[6]: # Crear una lista normal de python  
v1 = [2,4,6]  
v11 = [1,2,3]
```

```
[7]: # Crear un arreglo o vector de numpy  
v2 = np.array([2,4,6])  
v22 = np.array([1,2,3])
```

```
[8]: type(v1)
```

```
[8]: list
```

```
[9]: type(v2)
```

```
[9]: numpy.ndarray
```

```
[10]: v1 + v11
```

```
[10]: [2, 4, 6, 1, 2, 3]
```

```
[11]: v2 + v22
```

```
[11]: array([3, 6, 9])
```

```
[13]: # Crear un vector con unos, del tamaño que yo le indique en el argumento  
v3 = np.ones(3)  
v3
```

```
[13]: array([1., 1., 1.])
```

```
[14]: # Crear un vector dentro de un rango especificado (inicio, fin(hasta antes de))  
v4 = np.arange(1,8)  
v4
```

```
[14]: array([1, 2, 3, 4, 5, 6, 7])
```

```
[15]: v44 = np.array([1,2,3,4,5,6,7])  
v44
```

```
[15]: array([1, 2, 3, 4, 5, 6, 7])
```

```
[16]: # Multiplicación por escalar  
  
5 * v4
```

```
[16]: array([ 5, 10, 15, 20, 25, 30, 35])
```

```
[17]: 5 * v1
```

```
[17]: [2, 4, 6, 2, 4, 6, 2, 4, 6, 2, 4, 6, 2, 4, 6]
```

```
[18]: x = np.arange(1,5)  
y = np.array([2,4,6,8])
```

```
[19]: # Calculo del producto punto  
sum(x*y)
```

```
[19]: 60
```

```
[20]: # Calculo del producto punto con numpy  
np.dot(x,y)
```

```
[20]: 60
```

```
[21]: # Vectores Ortogonales o perpendiculares  
v1 = np.array([3,4])  
v2 = np.array([4,-3])
```

```
[22]: np.dot(v1,v2)
```

```
[22]: 0
```

```
[23]: # Calculo de la norma con numpy  
np.linalg.norm(x)
```

```
[23]: 5.477225575051661
```

```
[24]: np.linalg.norm(v1)
```

```
[24]: 5.0
```

```
[27]: # Creando una matriz en python  
A = np.array([[1, 3, 2],  
              [1, 0, 0],  
              [1, 2, 2]])
```

```
[26]: A
```

```
[26]: array([[1, 3, 2],  
          [1, 0, 0],  
          [1, 2, 2]])
```

```
[28]: B = np.array([[1, 0, 5],[7, 5, 0],[2, 1, 1]])
```

```
[29]: B
```

```
[29]: array([[1, 0, 5],  
          [7, 5, 0],  
          [2, 1, 1]])
```

```
[30]: # Multiplicación de matriz por escalar  
A * 2
```

```
[30]: array([[2, 6, 4],  
          [2, 0, 0],  
          [2, 4, 4]])
```

```
[31]: # Suma de matrices  
A + B
```

```
[31]: array([[2, 3, 7],  
          [8, 5, 0],  
          [3, 3, 3]])
```

```
[34]: # Resta de matrices  
A - B
```

```
[34]: array([[ 0,  3, -3],  
          [-6, -5,  0],  
          [-1,  1,  1]])
```

```
[35]: # La dimensión de la matriz  
A.shape
```

```
[35]: (3, 3)
```

```
[36]: # La cantidad de elementos de la matriz  
A.size
```

```
[36]: 9
```

```
[37]: # Ejemplo de multiplicacion de matrices  
A1 = np.arange(1, 13).reshape(3, 4)  
A1
```

```
[37]: array([[ 1,  2,  3,  4],
           [ 5,  6,  7,  8],
           [ 9, 10, 11, 12]])
```

```
[38]: B1 = np.arange(8).reshape(4,2)
      B1
```

```
[38]: array([[0, 1],
           [2, 3],
           [4, 5],
           [6, 7]])
```

```
[39]: A1 @ B1
```

```
[39]: array([[ 40,  50],
           [ 88, 114],
           [136, 178]])
```

```
[40]: B1 @ A1
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[40], line 1
----> 1 B1 @ A1
```

```
ValueError: matmul: Input operand 1 has a mismatch in its core dimension 0, with
signature (n?,k),(k,m?)->(n?,m?) (size 3 is different from 2)
```

```
[41]: # Creando matriz identidad
      I = np.eye(2)
      I
```

```
[41]: array([[1., 0.],
           [0., 1.]])
```

```
[42]: A2 = np.array([[4,7],[2,6]])
```

```
[44]: # Multiplicación de matriz identidad
      I @ A2
```

```
[44]: array([[4., 7.],
           [2., 6.]])
```

```
[45]: # Calcular el determinante
      np.linalg.det(A2)
```

```
[45]: 10.000000000000002
```

```
[46]: np.linalg.det(A)
```

```
[46]: -2.0
```

```
[49]: # Calcular la inversa  
A_inv = np.linalg.inv(A2)  
A_inv
```

```
[49]: array([[ 0.6, -0.7],  
          [-0.2,  0.4]])
```

```
[50]: A8 = np.array([[1,1,0],[2,-1,2],[0,3,0]])  
A8_inv = np.linalg.inv(A8)  
A8_inv
```

```
[50]: array([[ 1.          ,  0.          , -0.33333333],  
          [ 0.          ,  0.          ,  0.33333333],  
          [-1.          ,  0.5         ,  0.5         ]])
```

```
[ ]:
```